

SKETCH OF THE SOCIAL FOLDS THEORY

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ABSTRACT

This sketch proposes the Social Folds Theory as a heuristic to restore rigor to the social sciences. Facing the Anthropocene and the logarithmic scaling of hybrids, classical boundaries between nature, culture, and the social implode. Revisiting Durkheim, Marx, and Weber, we locate in Weber's hiatus - irrationalis - when focusing the lens blurs the whole - the fracture where folds emerge. Drawing on Simmel's sociology of forms [secret, conflict, the stranger] and Pareto's non-logical action [residues and derivations], we define social folds as variational events. Like a second-degree equation admitting x_1 and x_2 simultaneously, the social field sustains two possibles; probability operates as the fold operator. The agent does not choose the "true" meaning, but the one that spends less meaning-energy. Social folds are therefore a variational principle where interactions generate percussive waves that fold time-space and displace the sign.

KEYWORDS: Social Folds Theory; Variational Principle; Meaning-Energy.

There is a need to recover rigor in what is called "social sciences": "sciences of interaction", "sciences of macro-social processes", "sciences of structuration", to name a few (Giddens, 1984). The use in recent decades of more transversal cuts, highly justifiable in the folding, in the possibles and in the probables, radicalized the classical boundaries between culture, nature and the social, which are collated as signifier and signified of these poles. In this, interdisciplinary complexes become evident. What I want to present is that the complex problems that part of humanity is interested in solving, or at least mitigating, have signifier and signified; the sign implodes, like a nuclear fissure: the geological era - the Anthropocene - which unfolds through highly complex folds; we have mechanical difficulty finding ballast for radicality. New hybrids swarm, both in the cultural and in what we call natural, putting into question Enlightenment conceptions of what was perceived as modernity (Latour, 1991; Descola, 2005; Lévi-Strauss, 1962; Morin, 1977-2004).

Modernity was read by our thinkers as a necessary historical process because it was based on technological, technical, and dominance structures in the field of social interactions at micro and macro levels. On the other hand, modernity extends beyond Western Europe and takes on highly complex regional contours and in kinetic movement bordered by cultural variants - behavioral, linguistic, mystical-religious, and beyond - and informs how capitalism has a capillarity of amalgamating intersubjectivities and collectives that goes beyond the material and introjects itself into metabolism and electromagnetic waves. The colonies, in emerging modernity, did not come out unscathed: they felt and reacted to these foldings in diverse ways, and economic practices in zones of intense friction solidified, whether in the Americas, Asia, Africa, Oceania, the Middle East, Southeast Asia, among others. It is impossible to frame the photograph and capture the instant in which

capitalism merges with modernity: it seems more plausible to acquiesce that they are faces of the same coin, which does not stop spinning on the table of chance.

Equipped with a distinct heuristic proposal and epistemologically situated, in a historical instant still based on Mechanical Physics, Durkheim observes the instability of the urban-industrial European society, wisely maps anomie as condition and effect of social representations whose solidarities run the risk of extinguishing social integration (Durkheim, 2007 [1893]); Marx diagnoses the contradictions of the urban-industrial society rooted in the capitalist production model and inverts Hegelian idealism in favor of production relations that give meaning and capacity for overcoming from the material production of life (Marx, 2008 [1867]). Weber, on the other hand, proposes to understand the historical process that would ground the social despite the atomization of the social corpus, taking social action as the infinitesimal link that, in bundles of relations, gives organicity to institutions, modalities of domination, production, and economic development in diverse cultures to understand how elective affinities, social relations, and social processes framed in the historical context operate. This author, wisely, offers us the cliff to be presented to social folds: The meanings that social agents attribute are all complex, variable, intertwined in possibles and probables that, to guarantee methodological rigor, it is necessary to elect which intentionality is proposed to others and analyze it. But when this methodological undertaking is carried out, not only is an episteme created; it destabilizes the stability of the concept (ideal type). Weber himself assumes that capturing the meaning of action is impaired because, when focusing the lens towards something, when opening and closing the diaphragm of the photographic apparatus, the whole becomes out of focus. In this sense, I oppose Luhmann's postulate according to which social systems can control uncertainties in the participants' behavior (Luhmann, 2016). On the contrary. It is in this analytical fracture that Weber refers to *hiatus irrationalis*; there is always, in the possible, the possibility of a probable folding and undressing of the falsifiability of the concept. Or, as Giddens (1990) would say, where there are two, there is no certainty. It is in this fracture that I propose the concept of social folds, in the plural, because the object is no longer one, but multiple, constant, fluid, fractal. Reality can no longer be observed only by ocularity, which observes the object with flagrant optical limitations. It is necessary to institute new technical and technological models, using abstract systems in our favor.

This same reality, from now on, I will use the term in the plural, is no longer possible to investigate while ignoring the logarithmic scaling of hybrids, and I owe this concept to Latour (1991) and, for honesty, I intend to use it to control the conceptual hybrids that multiply at sonic speed and create more opacities than devices capable of measuring and locating the possibles and probables.

Simmel (1992 [1908]) wrote an effective theoretical sketch on the sociology of form. Individuals, for him, interact in a framework that multiplies, where part and whole intertwine; this is the perceptible schematic part that I call fold. Social forms are visualized as secret, conflict, the different-other. Pareto (1988 [1916]) provides us with the social as possible calculation. In this calculation, social action is not purely logical, walking and confronting itself with residues.

In a second-degree equation, Bhaskara gives us x_1 and x_2 . The same equation admits two possibles. Why would the social have to interpret only one to the detriment of the other? Probability is the fold operator. The social fold is precisely this: the coexistence of x_1 and x_2 in the same field of meanings. The agent does not choose the "true"; he chooses the one that spends less meaning-energy in that context. For example: to calculate the possibles

of the action of a social agent, where he can factorially choose the changing alternatives A, B, C, D, E (infinite), however, has available, on a barely visible scale, the choosable variables F, G, H, I, J (infinite) and, on a macro-scale, AB, CD, EF, GH, IJ, LM (infinite) understood as combinations of choices at micro and meso levels, what is the most adequate algebra to sketch this type of calculation?

I can present multilinear algebra. If choices at three levels occur simultaneously or mutually influence each other, you can treat each scale as an independent vector space, where:

Space V1 (Principal Scale): Dimension 5 (Base: {A, B, C, D, E})

Space V2 (Low-visibility Scale): Dimension 5 (Base: {F, G, H, I, J})

Space V3 (Macro-scale): Dimension N (Base: {AB, CD, EF, ...})

The space of all possible states is the tensor product of these spaces:

$$V_{\text{total}} = V_1 \otimes V_2 \otimes V_3$$

A specific state would be a tensor like $A \otimes F \otimes AB$, which represents choice A at the principal scale, under hidden condition F, within macro scenario AB. If there are probabilities or weights associated, linear combinations of these tensors are used.

To answer the question using Multilinear Algebra, one must first map the logical structure of variables and understand how tensors operate in this scenario. In pure tensor algebra, the direct answer is: the possibility of agent X1 deciding in isolation for X9 is equal to zero (or null), because he is operating in the fold of possibles and not probables. This happens because X9 belongs to the base of a vector space (barely visible scale) completely different from that where the agent's direct choice options are (X2, X3, X4). In multilinear algebra, an agent does not choose an isolated element from another scale; he chooses a combined state (a tensor) that unites the three scales. X9 is not an action in itself, but a hidden boundary condition that affects the action.

Below, we demonstrate mathematically how to calculate the probability of X9 being part of the final decision of agent X1.

1. Modeling of Tensor Spaces

Let's define the three scales as independent vector spaces:

Space V (Direct Choice / Visible Scale): Dimension 3. Base: {e_{X2}, e_{X3}, e_{X4}} (the three target agents of interest).

Space W (Low-visibility Scale / Hidden Dynamics): Dimension 5. Base: {e_{X5}, e_{X6}, e_{X7}, e_{X8}, e_{X9}} (underlying factors, e.g., hidden feelings, veiled social pressures).

Space M (Macro-scale / Social Context): Dimension 6 (limiting your example "..."). Base: {e_{2x}, e_{3x}, e_{4x}, e_{5x}, e_{6x}, e_{7x}} (e.g., cultural norms, contact network configurations).

The space of all possible realities for X1's action is the tensor product:

$$T = V \otimes W \otimes M \text{ (Dimension } 3 \cdot 5 \cdot 6 = 90 \text{ possible states)}$$

2. Weight Assignment (Example of State Vectors)

To perform the calculation, we need to define X1's propensities and conditions:

Direct Choice Vector (v): The agent is torn. He maintains 60% attachment to his girlfriend (X2), but has 20% interest in X3 and 20% in X4.

$$v = 0.6 e_{\{X2\}} + 0.2 e_{\{X3\}} + 0.2 e_{\{X4\}}$$

Hidden Scale Vector (w): Here, variable X9 enters. Let's say the environment or psychological state of the agent activates factor X9 with 30% intensity, distributing the rest among other hidden variables.

$$w = 0.1 e_{\{X5\}} + 0.2 e_{\{X6\}} + 0.2 e_{\{X7\}} + 0.2 e_{\{X8\}} + 0.3 e_{\{X9\}}$$

Macro-scale Vector (m): The macro scenario of the monogamous regime imposes strong pressure (Example: norm 2x with 100% weight).

$$m = 1.0 e_{\{2x\}}$$

3. The Global Tensor Product Calculation

The state of X1's action is given by the equation: $T = v \otimes w \otimes m$

Substituting the vectors and calculating the distributive distribution only for the terms that activate X9:

$$T_{\{with\ X9\}} = (0.6 e_{\{X2\}} + 0.2 e_{\{X3\}} + 0.2 e_{\{X4\}}) \otimes (0.3 e_{\{X9\}}) \otimes (1.0 e_{\{2x\}})$$

Expanding this fraction of the tensor, we obtain the 3 specific scenarios where X9 coexists with the agent's decisions:

$$T_{\{with\ X9\}} = 0.18(e_{\{X2\}} \otimes e_{\{X9\}} \otimes e_{\{2x\}}) + 0.06(e_{\{X3\}} \otimes e_{\{X9\}} \otimes e_{\{2x\}}) + 0.06(e_{\{X4\}} \otimes e_{\{X9\}} \otimes e_{\{2x\}})$$

Conclusion: The possibility of X9 being present and active in X1's decision dynamics is the sum of coefficients where it appears ($0.18+0.06+0.06 = 0.30$, i.e., 30% global chance of hidden factor X9 manifesting). However, feasible decisions of X1 under the influence of X9 divide into three paths: maintaining girlfriend X2 but under the veiled effect of X9: 18% chance. Choosing agent X3, hiddenly driven by X9: 6% chance. Choosing agent X4, hiddenly driven by X9: 6% chance. In short, X1 will never "decide for X9", but multilinear algebra proves there is a 12% chance ($6\%+6\%$) of hidden factor X9 making the agent break monogamy and choose one of the other suitors (X3 or X4).

Calculation focused on X3

To calculate the possibility of social agent X1 deciding specifically for X3, we need to analyze the geometric structure of the tensor product. The tensor vector is on a smaller scale. The probability of choosing X3 does not depend only on isolated desire, but on how this choice propagates through all layers.

Adopting the same realistic probabilistic weights:

$$v = 0.6 e_{\{X2\}} + 0.2 e_{\{X3\}} + 0.2 e_{\{X4\}}$$

$$w = 0.2 e_{\{X5\}} + 0.2 e_{\{X6\}} + 0.2 e_{\{X7\}} + 0.2 e_{\{X8\}} + 0.2 e_{\{X9\}}$$

$$m = 1.0 e_{\{2x\}}$$

Isolating term $0.2 e_{X3}$:

$$T_{\{X3\}} = (0.2 e_{\{X3\}}) \otimes (0.2 e_{\{X5\}} + 0.2 e_{\{X6\}} + 0.2 e_{\{X7\}} + 0.2 e_{\{X8\}} + 0.2 e_{\{X9\}}) \otimes (1.0 e_{\{2x\}})$$

Expanding the 5 scenarios where X1 migrates his decision to X3:

$$\text{With hidden factor X5: } 0.2 \cdot 0.2 \cdot 1.0 = 0.04(e_{X3} \otimes e_{X5} \otimes e_{2x})$$

$$\text{With hidden factor X6: } 0.04(e_{X3} \otimes e_{X6} \otimes e_{2x})$$

$$\text{With hidden factor X7: } 0.04(e_{X3} \otimes e_{X7} \otimes e_{2x})$$

$$\text{With hidden factor X8: } 0.04(e_{X3} \otimes e_{X8} \otimes e_{2x})$$

With hidden factor X9: $0.04(e_{x3} \otimes e_{x9} \otimes e_{2x})$

$T_{\{X3\}} = 0.04_{\{X5\}} + 0.04_{\{X6\}} + 0.04_{\{X7\}} + 0.04_{\{X8\}} + 0.04_{\{X9\}}$

For the scalar final probability:

$P(X_3) = 0.04 + 0.04 + 0.04 + 0.04 + 0.04 = \mathbf{0.20 (20\%)}$

Sociological interpretation: The global possibility of X1 deciding for X3 is 20%. This probability is stratified. It depends on 5 different hidden micro-contexts operating under the same macro-structural pressure (2x). Each specific micro-context carries exactly a 4% chance. If an external event alters the macro-scale, tensor math will instantly recalculate the viability of X3 under new boundary conditions.

Thus, social folds, in the plural, are events in which the social agent establishes choices through forms of meaning and elects the one that spends less meaning-energy. This meaning-energy is multiform and combines as a binomial system, generating events not always predicted. Hence the challenge of understanding historical events as an internally cohesive picture: cultures traverse everyone and psychobiological energy, animated by Physics, does not coalesce in the same direction. Social folds are a variational principle in which social agents interact, generating events that overflow in percussive waves, whose effects are only partially perceived, and the event folds time-space, crushing the sign which, when displaced, can, in the binomial universe of senses, exchange position.

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